

Code-Layout, Readability and Reusability

Date	12 JULY 2026
Team ID	N/A - Individual Submission
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python and JSX files use consistent indentation.
2	Proper File Structure	Files and folders are logically organized	Yes	backend/app routers, schemas, models and a utilities are separated; frontend components are organized.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names like monthly_income, debt_to_income, predict_settlement and generate_negotiation_letter are clear.
4	Function / Method Names	Functions are descriptively named	Yes	Functions describe their business purpose
5	Code Comments	Inline and block comments explain log	Partial	Code is mostly self-explanatory; comments are used sparingly.
6	Modular Design	Code is split into reusable functions/m	Yes	Auth, loans, analysis and settlement are separate routers.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared API helper and components avoid repeated request logic.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Frontend displays API errors; backend validates inputs through Pydantic

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	apiRequest	JavaScript	Dashboard, Settlement and Letter API calls	High

2	Financial scoring utilities	Python	analysis endpoint and documentation logic	High
3	SQLModel models	Python	Routers and database initialization	High
4	Dashboard Field component	React	Financial and loan input fields	Medium
5	Docker configuration	Docker	Local and evaluator deployment	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure		
Readability		
Reusability		
Documentation / Comments		
Overall Score		